SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 1.2: Indices and Logarithms — Teacher Reference (Pre-filled)

SUMMARY TABLE: CORE MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 1.2: Indices and Logarithms
Driving Question	DRIVING QUESTION: How can a Kericho tea farmer — or any Kenyan scientist, engineer, or business person — use the mathematics of indices and logarithms to make sense of, and calculate with, numbers that are too large or too small to handle by ordinary arithmetic?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Reciprocals and the Language of Indices: Launching the Tea Farm Phenomenon**	Students examined the Kericho tea-farm phenomenon: a farmer tracking exponential growth of tea-leaf output, bacterial contamination levels in processed tea, and loan repayment schedules — all involving numbers too large or too small for ordinary arithmetic. They observed that repeated multiplication produces rapidly growing (or shrinking) quantities and that writing these out in full quickly becomes impractical. They recorded examples such as 10 × 10 × 10 × 10 × 10 and compared the full expanded form with the compact index notation 10^5.	Students learned two foundational ideas: (1) The reciprocal of any non-zero real number a is 1/a, and the defining property is a × (1/a) = 1 — every number has exactly one multiplicative inverse. (2) Index (exponential) notation a^n means 'multiply a by itself n times'; a is the base, n is the index (exponent), and the notation compresses repeated multiplication into a single compact expression. Pedagogical note: Stress that the reciprocal relationship will reappear in Lesson 3 as the negative-index rule, so building fluency here pays dividends later.	Students explained why a Kericho tea farmer — or any scientist or business person — needs a compact notation: numbers like the mass of a single tea-leaf cell ($\sim 10^{-12}$ g) or Kenya's annual tea export revenue ($\sim 10^{10}$ KES) are unworkable in expanded form. Index notation is not mere shorthand; it encodes the multiplicative structure that makes the Laws of Indices (Lessons 2–3) and logarithms (Lessons 4–8) possible. Students connected reciprocals to the idea of 'undoing' multiplication, previewing the inverse relationship between indices and logarithms.
Lesson 2 **Deriving the Laws of Indices — Part 1: Product Rule, Quotient Rule, and Zero Index Law**	Students expanded expressions such as ($10^2 \times 10^3$) and ($2^5 \div 2^2$) by writing out every factor explicitly, counted the resulting factors, and recorded the pattern. They tabulated multiple examples — using bases relevant to the tea-farm context (e.g., powers of 2 for doubling bacterial colonies, powers of 10 for currency and mass) — until the general pattern became visible across all cases, including the special case $a^n \div a^n$.	Students derived and can now state three laws: (1) Product Rule — $a^m \times a^n = a^{m+n}$: when multiplying powers of the same base, add the exponents. (2) Quotient Rule — $a^m \div a^n = a^{m-n}$ ($a \neq 0$): when dividing powers of the same base, subtract the exponents. (3) Zero Index Law — $a^0 = 1$ ($a \neq 0$): derived as the logical consequence of the quotient rule when m = n. Pedagogical note: Insist students derive, not memorise — the ability to re-derive a law from first principles is the	Students explained each law in the tea-farm context: the product rule models combining two growth stages (e.g., tea-leaf area doubling over 2 weeks then 3 more weeks = $2^2 \times 2^3 = 2^5$); the quotient rule models reducing or scaling back (e.g., dividing a large batch by a sub-batch); the zero index law resolves the apparently paradoxical result that any non-zero quantity divided by itself gives 1, consistent with $a^0 = 1$. Students updated the class Driving

		competency assessed in CBE performance tasks.	Question Board, noting that these three laws are the algebraic engine behind logarithm laws in later lessons.
Lesson 3 **Deriving the Laws of Indices — Part 2: Power, Negative, and Fractional Indices**	Students applied the product rule repeatedly to $(a^2)^3 = a^2 \times a^2 \times a^2$, counted factors, and generalised. They used the quotient rule to evaluate $a^0 \div a^n$ and discovered the negative index. They explored expressions like $4^{(1/2)}$ by asking 'what number, multiplied by itself, gives 4?' and connected this to square roots, extending to cube roots and general fractional indices with a Kenyan context (e.g., the side length of a square tea plot of area $2^{(1)}$ hectares).	Students learned and can state three further laws: (1) Power Rule — $(a^m)^n = a^{mn}$: raise a power to a power by multiplying exponents. (2) Negative Index Rule — $a^{-n} = 1/a^n$ ($a \neq 0$): a negative exponent denotes the reciprocal of the positive-exponent form, directly linking back to Lesson 1's reciprocal concept. (3) Fractional Index Rule — $a^{(1/n)} = \sqrt[n]{a}$ and $a^{(m/n)} = (\sqrt[n]{a})^m = \sqrt[n]{a^m}$: a fractional exponent denotes a root, with the denominator indicating which root and the numerator indicating the power. Pedagogical note: The negative-index and fractional-index rules are the most commonly misapplied; plan for targeted error analysis in Lesson 8's consolidation.	Students explained how all six laws of indices form a coherent system: the product rule generates the power rule; the quotient rule generates the zero and negative index rules; the fractional index rule generalises the idea of roots. In the Kericho tea-farm context, students explained why a scientist computing the geometric mean of bacterial counts (a fractional index operation) or expressing a very small pesticide concentration in scientific notation (negative index) is using these same rules. The model was updated to show indices as a complete, rule-governed language for exponential quantities.
Lesson 4 **Introduction to Common Logarithms: Definition, Characteristic, Mantissa, and Reading Four-Figure Tables**	Students examined a printed four-figure logarithm table (the type used in Kenyan secondary schools before calculators) and the pattern of powers of 10: $10^0 = 1$, $10^1 = 10$, $10^2 = 100$, $10^3 = 1\,000$, etc. They observed that numbers between consecutive powers of 10 have logarithms between consecutive whole numbers and that the table encodes these fractional parts. They practised locating values such as log 2.34 and log 7.89, noting the row-and-column structure and the difference columns for interpolation.	Students learned the formal definition: $\log_{10} x = n$ if and only if $10^n = x$ ($x > 0$). They learned to decompose any common logarithm into (1) the characteristic — the integer part, determined by the number of digits to the left of the decimal point (characteristic = number of digits − 1 for numbers ≥ 1) — and (2) the mantissa — the non-negative decimal part, always read from the four-figure table using the first four significant figures of the number. They practised writing any positive number in standard form (scientific notation) as a bridge to finding the characteristic. Pedagogical note: Many students confuse the characteristic rule for numbers less than 1 (negative characteristics written using bar notation, e.g., $\bar{2}$); allow extra practice time here before Lesson 5.	Students explained the logarithm as the answer to the question: 'To what power must 10 be raised to give this number?' They connected this directly to the tea-farm driving question: a farmer's annual yield of 3 500 000 kg has $\log_{10}(3\,500\,000) \approx 6.544$, meaning it sits between 10^6 and 10^7, and the mantissa 0.544 locates it precisely within that interval. The four-figure table is the farmer's (or scientist's) practical tool for finding that mantissa without a calculator. Students updated the DQB: 'The logarithm translates a multiplication problem into an addition problem — but how?'
Lesson 5 **Antilogarithms: Reversing the Logarithm and Using Four-Figure Antilog Tables**	Students observed that finding a logarithm is only half the calculation — the final answer must be converted back to an ordinary number. They examined the antilogarithm (antilog) table, compared its structure with the log table, and traced a complete round-trip: starting with a number,	Students learned that the antilogarithm is the inverse operation of the logarithm: if $\log_{10} x = n$, then $x = \text{antilog}(n) = 10^n$. They learned the procedure for reading the antilog table: (1) use only the mantissa (decimal part) to look up the four-figure antilog value, and (2) use the characteristic	Students explained the antilogarithm as the bridge that makes logarithm-based calculation complete and useful. In the Kericho context: a farmer uses the log table to turn a difficult multiplication into addition, performs the addition, and then uses the antilog table to translate the result back into

	finding its log, then using the antilog table to recover the original number. They used numbers drawn from the tea-farm context (e.g., a computed log result of 4.7193) and verified recovery by checking against a known power of 10.	to place the decimal point in the final answer (characteristic + 1 gives the number of digits before the decimal point for positive characteristics). They practised mixed sets of log and antilog table readings, including cases with negative (bar-notation) characteristics. Pedagogical note: Reinforce that the antilog table is simply a table of powers of 10; students who understand the definition from Lesson 4 should be able to reconstruct the procedure from first principles.	kilograms, shillings, or litres. Students articulated the full two-step workflow (log → arithmetic → antilog) that will underpin Lessons 6–8, and updated the DQB with the insight: 'Logs convert × into +, and antilogs convert the answer back — so difficult multiplication becomes simple addition.'
Lesson 6 **Applying Logarithms to Multiplication and Division**	Students observed the derivation of the product law from the index law: if a = 10^(log a) and b = 10^(log b), then ab = 10^(log a + log b), so log(ab) = log a + log b. They worked through a fully tabulated four-figure-table calculation — multiplying two large numbers relevant to the tea farm (e.g., 3 847 × 2 163, representing kilograms times price per kilogram) — recording each step: find log of each factor, add the logs, find the antilog. They repeated the process for division using the quotient law log(a/b) = log a − log b.	Students derived and applied two logarithm laws: (1) Product Law — log(ab) = log a + log b: logarithm of a product equals the sum of the logarithms. (2) Quotient Law — log(a/b) = log a − log b: logarithm of a quotient equals the difference of the logarithms. Students practised multi-factor multiplications (three or more factors) by extending the product law, and divisions involving numbers less than 1 (requiring bar-notation arithmetic). Pedagogical note: Emphasise that these laws are not new rules to memorise but direct consequences of the index laws derived in Lessons 2–3; students who see this connection develop more robust, transferable understanding.	Students explained why logarithms were historically the essential computational tool for Kenyan agricultural scientists, engineers, and accountants before electronic calculators: they convert the hardest arithmetic operations (multiplication and division of large or small numbers) into the simplest (addition and subtraction). They connected the product law back to the product rule of indices and the quotient law back to the quotient rule, completing the conceptual arc that began in Lesson 2. The DQB was updated: 'The log laws are the index laws in disguise — multiplication of numbers becomes addition of exponents.'
Lesson 7 **Logarithms of Powers and Roots: The Power Law and Root Law**	Students observed the derivation of the power law: log(a^n) = log(a × a × … × a) = log a + log a + … + log a (n times) = n log a, connecting directly to the product law of Lesson 6 and the power rule of indices from Lesson 3. They worked through calculations involving squares, cubes, and roots using four-figure tables — for example, computing the cube root of a large tea-yield figure (e.g., $\sqrt[3]{(2\ 197\ 000\ \text{kg}^3)}$ to find a geometric mean) — and verified results by squaring or cubing back.	Students derived and applied two further logarithm laws: (1) Power Law — log(a^n) = n log a: to find the log of a power, multiply the log of the base by the exponent. (2) Root Law — log($\sqrt[n]{a}$) = (1/n) log a: to find the log of a root, divide the log of the number by the index of the root (a direct consequence of the fractional index rule from Lesson 3). Students practised calculations involving non-integer and fractional powers, including cases where the bar-notation characteristic must be divided carefully (e.g., $\bar{3}.6020$ ÷ 3). Pedagogical note: The division of a bar-notation characteristic is a frequent source of error; model the adjustment step explicitly — e.g., rewrite $\bar{3}.6020$ as (−3 + 0.6020) and divide each part separately.	Students explained the power law and root law as completing the logarithm toolkit: together with the product and quotient laws from Lesson 6, these four laws allow any combination of multiplication, division, powers, and roots to be handled by addition, subtraction, multiplication, and division of logarithms. In the Kericho tea-farm context, students showed how a scientist computing the geometric mean yield across several farms, or an engineer computing a root of a very large pressure value, applies these laws in a single systematic table-based calculation. The model diagram on the DQB was extended to include powers and roots.

Lesson 8 **Consolidation, Combined Operations, and Model Revision: Answering the Driving Question**	Students worked through multi-step expressions combining all four logarithm laws — product, quotient, power, and root — in a single calculation, using four-figure log and antilog tables throughout. Problems were drawn directly from the Kericho tea-farm driving question context (e.g., computing compound interest on a farm loan, estimating bacterial growth rates, converting very small pesticide concentrations). Students also reviewed and corrected common errors identified across Lessons 1–7 — particularly negative-index misapplication, bar-notation arithmetic, and characteristic-placement errors — through structured peer error analysis.	Students consolidated the complete conceptual map of the sub-strand: (1) all six laws of indices form a coherent algebraic system; (2) logarithms are the inverse of indices and inherit their laws in translated form; (3) the four logarithm laws — product, quotient, power, root — together enable calculation with any combination of large/small numbers using only four-figure tables and basic arithmetic; (4) any multi-step expression can be solved by planning the sequence of log-law applications before computing. Students articulated the unity of the sub-strand: indices and logarithms are two perspectives on the same exponential structure. Pedagogical note: Use this lesson's error-analysis activity diagnostically — persistent errors in bar-notation arithmetic or characteristic placement indicate gaps in Lesson 4 or 7 that may need targeted follow-up before formal assessment.	Students returned to the driving question and answered it fully: a Kericho tea farmer, Kenyan scientist, engineer, or business person uses indices to express and manipulate very large or very small quantities compactly and accurately, and uses logarithms to convert multiplicative problems into additive ones — making otherwise intractable calculations manageable with nothing more than a four-figure table and careful arithmetic. Students revised their cumulative model to show the complete pathway: real-world exponential quantity → standard form (indices) → logarithm (log table) → arithmetic with log laws → antilogarithm (antilog table) → real-world answer. The Driving Question Board was marked complete.